

Step-by-Step Design and Analysis of a PMOS LDO Using a 2-Stage Miller Compensated OTA

1 Introduction

This document presents the complete step-by-step design and analysis of a Low Dropout Regulator (LDO) using a two-stage Miller compensated Operational Transconductance Amplifier (OTA) in a 130 nm CMOS process. The design covers OTA gain analysis, frequency compensation, pass transistor sizing, stability considerations, and regulation performance.

2 Two-Stage OTA Design

2.1 First Stage Gain

From operating-point simulation:

$$g_{m1} \approx 120 \mu\text{S} \quad (1)$$

$$r_{o1} \approx 65 \text{ k}\Omega \quad (2)$$

The first-stage gain is:

$$A_1 = g_{m1}r_{o1} = 120 \times 10^{-6} \times 65 \times 10^3 \approx 7.8 \quad (3)$$

2.2 Second Stage Gain

From simulation:

$$g_{m2} \approx 1.1 \text{ mS} \quad (4)$$

$$r_{o2} \approx 40 \text{ k}\Omega \quad (5)$$

The second-stage gain is:

$$A_2 = g_{m2}r_{o2} = 1.1 \times 10^{-3} \times 40 \times 10^3 \approx 44 \quad (6)$$

2.3 Overall OTA Gain

$$A_{\text{OTA}} = A_1 \times A_2 \approx 7.8 \times 44 \approx 343 (\approx 51 \text{ dB}) \quad (7)$$

This gain is sufficient to ensure good DC regulation accuracy.

3 Miller Compensation Analysis

3.1 Dominant Pole

Given Miller capacitor:

$$C_c = 2.7 \text{ pF}$$

The dominant pole is approximated by:

$$f_{p1} \approx \frac{1}{2\pi r_{o1} C_c (1 + A_2)} \quad (8)$$

Substituting values:

$$f_{p1} \approx \frac{1}{2\pi \times 65k \times 2.7p \times 45} \approx 20 \text{ kHz} \quad (9)$$

3.2 Zero-Nulling Resistor

For proper zero placement:

$$R_z \approx \frac{1}{g_{m2}} \approx \frac{1}{1.1m} \approx 900 \Omega \quad (10)$$

The resulting zero location:

$$f_z = \frac{1}{2\pi R_z C_c} \approx 65 \text{ MHz} \quad (11)$$

This zero cancels the second-stage pole.

4 PMOS Pass Transistor Design

4.1 Dropout Requirement

Given:

$$I_{L,\max} = 50 \text{ mA} \quad (12)$$

$$V_{\text{drop}} = 200 \text{ mV} \quad (13)$$

$$R_{\text{on},\max} = \frac{V_{\text{drop}}}{I_L} = \frac{0.2}{0.05} = 4 \Omega \quad (14)$$

4.2 Pass Device Sizing

In the linear region:

$$I_D \approx \mu_p C_{ox} \frac{W}{L} (V_{SG} - |V_T|) V_{SD} \quad (15)$$

Assuming:

$$\mu_p C_{ox} \approx 50 \mu\text{A}/\text{V}^2 \quad (16)$$

$$V_{SG} - |V_T| \approx 0.3 \text{ V} \quad (17)$$

$$V_{SD} = 0.2 \text{ V} \quad (18)$$

$$\frac{W}{L} \approx \frac{50m}{50\mu \times 0.3 \times 0.2} \approx 1.67 \times 10^4 \quad (19)$$

Choosing:

$$L = 0.5 \mu\text{m} \quad \Rightarrow \quad W \approx 8 \text{ mm}$$

4.3 Gate Capacitance

$$C_{g,\text{pass}} \approx C_{ox}WL \approx 30 \text{ pF} \quad (20)$$

5 Feedback Network Design

The regulated output voltage is given by:

$$V_{out} = V_{ref} \left(1 + \frac{R_1}{R_2} \right) \quad (21)$$

Given:

$$V_{out} = 1.5 \text{ V}, \quad V_{ref} = 0.6 \text{ V}$$

$$\frac{R_1}{R_2} = \frac{1.5}{0.6} - 1 = 1.5 \quad (22)$$

Chosen values:

$$R_1 = 300 \text{ k}\Omega, \quad R_2 = 200 \text{ k}\Omega$$

Feedback factor:

$$\beta = \frac{R_2}{R_1 + R_2} = 0.4 \quad (23)$$

6 LDO Loop Dynamics

6.1 Output Pole

At light load:

$$R_L = \frac{V_{out}}{I_L} = \frac{1.5}{1m} = 1.5 \text{ k}\Omega \quad (24)$$

With:

$$C_{out} = 1 \mu\text{F}$$

$$f_{out} = \frac{1}{2\pi R_L C_{out}} \approx 100 \text{ Hz} \quad (25)$$

6.2 ESR Zero

Assuming:

$$R_{ESR} = 0.1 \Omega$$

$$f_{z,ESR} = \frac{1}{2\pi R_{ESR} C_{out}} \approx 1.6 \text{ kHz} \quad (26)$$

6.3 Gate Pole

$$f_g = \frac{1}{2\pi R_{out,OTA} C_{g,pass}} \approx 120 \text{ kHz} \quad (27)$$

6.4 Unity Gain Bandwidth

For stability:

$$\text{UGB} \leq \frac{1}{3} f_g \quad (28)$$

$$\boxed{\text{UGB} \approx 50\text{--}100 \text{ kHz}} \quad (29)$$

7 Regulation Performance

7.1 Line Regulation

From DC sweep:

$$\Delta V_{out} = 1.9 \text{ mV} \quad (30)$$

$$\Delta V_{in} = 1.2 \text{ V} \quad (31)$$

$$\text{Line Regulation} = \frac{\Delta V_{out}}{\Delta V_{in}} \approx 1.58 \text{ mV/V} \quad (32)$$

7.2 Load Regulation

$$\Delta V_{out} \approx 1.8 \text{ mV} \quad (33)$$

$$\Delta I_L = 50 \text{ mA} \quad (34)$$

$$\text{Load Regulation} = \frac{1.8}{50} \approx 0.036 \text{ mV/mA} \quad (35)$$

8 Conclusion

A stable PMOS LDO has been successfully designed using a two-stage Miller compensated OTA. The regulator achieves good DC accuracy, acceptable line and load regulation, and robust stability through proper pole-zero placement and compensation.